

TCS349: Responsible and Explainable AI

Unit 2, Lecture 2

Transparency in AI: Meaning, Importance & Limitations

Why AI systems must be understandable, traceable and honest about what they can and cannot do - and how transparency is put into practice.

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What Transparency Means



Traceability



**Communicating
Limitations**



Stakeholders

2. What Does "Transparency" Mean in AI?

Transparency is the degree to which an AI system's workings, data and decisions can be seen, understood and questioned by relevant stakeholders.

It spans the full system: the data used, the model's logic, and the process by which the system was built and deployed.

Transparency is not "all or nothing" - it exists on a spectrum, from fully open-source to a closed black box.

A system can be transparent about its existence and use even if its internal logic is complex or proprietary.

3. Why Transparency Matters

Builds trust: users are more willing to rely on a system whose behaviour they can understand.

Enables contestability: people affected by a decision can only challenge it if they understand how it was made.

Supports regulatory compliance: many data-protection and AI laws require explanations for automated decisions.

Aids debugging and improvement: transparent systems are easier for developers to audit, test and correct.

4. Who Needs Transparency?

Developers and engineers: need visibility into model behaviour to debug, test and improve the system.

End users: need enough information to use the system appropriately and calibrate their trust in it.

Regulators and auditors: need evidence that the system complies with legal and ethical standards.

Affected individuals: need enough explanation to understand and, if necessary, contest a decision made about them.

5. Transparency vs Explainability

Dimension	Transparency	Explainability
Focus	The system as a whole - data, process, governance	A specific prediction or decision
Key Question	What is this system, and how was it built?	Why did the model produce this output?
Example	Publishing a model card describing purpose and limitations	Using SHAP to explain one rejected application
Relationship	The broader umbrella	One tool used to achieve transparency

6. Traceability in AI Systems - What It Means

Traceability is the ability to track an AI system's data, decisions and changes back to their origin.

It allows a specific output to be linked back to the exact data, model version and configuration that produced it.

Traceability turns a one-time transparency claim into something that can be verified after the fact.

Without traceability, "we can explain our system" is a claim that cannot actually be checked.

7. Traceability in Practice

Data lineage: recording where training data came from and how it was processed before use.

Model versioning: keeping a record of every trained model version and the changes between them.

Decision logs: storing the inputs, outputs and model version used for each individual decision.

Together these let an organisation reconstruct exactly why a system behaved a certain way at a certain time.

8. Communicating Model Limitations - Why It Matters

Every AI model has boundaries: conditions under which its predictions are less reliable or invalid.

Failing to communicate limitations can lead users to over-trust a system outside its intended scope.

Clear communication of limitations is part of transparency, not an admission of failure.

Regulators increasingly expect documented limitations alongside a system's capabilities.

9. How to Communicate Limitations Effectively

Use clear, non-technical language that a non-expert stakeholder can understand.

State known failure modes and the conditions under which the system should not be trusted.

Where possible, provide confidence levels or uncertainty estimates alongside predictions.

Clearly define the system's intended use, and explicitly flag unsupported or out-of-scope uses.

10. Levels of Transparency

Level	What It Covers
Algorithmic Transparency	The logic, features and process by which the model produces an output
Interaction Transparency	How the system communicates its behaviour and limitations to the user in the moment
Social Transparency	How the system's use, governance and impact are communicated to wider society

11. Barriers to Transparency

Trade secrets: organisations may withhold model details to protect competitive advantage.

Model complexity: some models are inherently difficult to describe in a way non-experts can understand.

Security concerns: full transparency can expose a system to manipulation or adversarial attacks.

Risk of over-simplification: an explanation simplified too far can mislead rather than inform.

12. Case Example: Transparency in Practice

A bank publishes a model card for its credit-scoring AI, describing its purpose, training data, key limitations and intended use.

The document lists demographics on which the model was validated and any groups for which performance is less certain.

Applicants receive a plain-language explanation whenever a decision affects them, alongside a way to request human review.

This single practice touches transparency, traceability and communicated limitations all at once.

13. Transparency Trade-offs

Transparency versus security: revealing too much about a system's logic can help bad actors game or attack it.

Transparency versus intellectual property: full disclosure can undermine a company's competitive position.

Transparency versus privacy: explaining a decision in detail can sometimes reveal sensitive information about other individuals.

Responsible AI treats these as trade-offs to be managed deliberately, not reasons to avoid transparency altogether.